Errors Are Not Choke Points: Difficulty, Error Onset and Counterfactual Commitment in LLM Reasoning Failures

Harsh Dubey Harshit Saini Prashant Jain
Himanshu Kumar Gaurav Kaushal Rohit Chaudhary
Dept. of CSE (AI&ML), KIET Group of Institutions, Ghaziabad, India
harsh.2428cseaiml127@kiet.edu

Abstract

Process supervision assumes that the first incorrect step of a reasoning trace is where it fails, and verifier-guided search backtracks to that step. We test this on ProcessBench's expert first-error labels.

Observationally, early-warning detectors mostly read problem difficulty: question and generator alone reach AUROC 0.588, and a quarter of the reasoning adds +0.005 (n.s.). 23.5% of first errors are not fatal.

We then measure commitment directly. We resample four open models from two families at every prefix of their own traces (125,832 scored continuations, up to 64 per step in independent batches) and show that the resulting value curves and decisive steps are reliable. 33–66% of failures are decided before the first step. When the outcome is still open, the labelled error is where success collapses in 0.438–0.647 of traces, against a null of about 0.18. This was confirmed in a preregistered test on Llama-3.1-8B.

For three Qwen models, resampling from just before the labelled error succeeds less often than restarting, and the effect replicates on independent samples (+0.140). It does not generalize to Llama, where the preregistered test fails (-0.022).

At a matched end-to-end budget that counts every token, no practical policy beats restart reliably. This includes backtracking guided by a validated process reward model (ProcessBench F1 73.2), which is worse than restart for Llama (-0.063). A short supermartingale argument explains why restart is hard to beat.

We release all rollouts, the preregistration and the code behind every number.

1 Introduction

Chain-of-thought (CoT) reasoning [1] turns a single prediction into a trajectory of steps, and two widely used ideas assume that a wrong final answer is decided at an identifiable point inside that trajectory. Process supervision [2, 3, 5] trains and evaluates verifiers to find the *first incorrect step*, and verifier-guided search backtracks to it. Early-warning detectors [9, 10] predict from a partial trajectory that the answer will be wrong, so that computation can be stopped or redirected.

Both ideas rest on two assumptions that are rarely tested. The first is that the signal they use is about the *trajectory*: a detector that reads a prefix should know more than a detector that reads only the question. The second is that the *annotated first error is the point at which the outcome is decided*. We call a step that decides the outcome a *choke point*, and ask whether errors and choke points coincide.

We test both assumptions on ProcessBench [5], which provides expert first-error labels for 3,400 solutions written by 12 open models. We then measure commitment directly: we resample four open models from two families at every prefix of their own traces, scoring 125,832 continuations. Every claim below carries its evidence level (Section 3): PRE (preregistered before any data), EXP (exploratory: found while looking at the data) or CONF (a preregistered test on new data). Our contributions:

1. **Difficulty-controlled evaluation of early failure detection** (EXP, CPU). A detector that reads only the question and the generator's identity reaches AUROC 0.588. Adding the first quarter of the reasoning adds +0.005 (n.s.), and within a problem the 25% prefix is at chance (0.485).
2. **Errors are not choke points, observationally** (EXP). 23.5% of first errors are followed by a correct answer, fatality does not depend on where the error occurs, and failure detectors rank non-fatal errors as riskier than fatal ones (AUROC 0.456).
3. **Reliable counterfactual value curves.** With up to 64 continuations per step, drawn as four independent batches, value curves replicate across batches ($r = 0.985$). Decisive steps are stable under posterior resampling, so disagreement with human labels is not sampling noise.
4. **Where failures are decided.** 33–66% of failures are capability-limited before the first step. When the outcome is still open, the annotated error is where success collapses in 0.438–0.647 of traces, against a position-matched null of 0.16–0.20 (PRE for the Qwen models; CONF on Llama, 0.633).
5. **Backtracking to the human first error is model-dependent.** For three Qwen models, resampling from just before the annotated first error is worse than restarting, by +0.068 to +0.165 per attempt (EXP). The effect replicates on independent samples (+0.140, CONF) but *fails to generalize* to Llama-3.1-8B (-0.022, CONF, not confirmed). A short supermartingale argument shows that restart is a strong baseline for any restart point chosen without looking

ahead.

6. **No practical policy beats restart at matched compute.** We count every token, including the cost of finding the restart point. Under that accounting, backtracking guided by a validated process reward model (ProcessBench F1 73.2) is worse than restart for Llama (-0.063, CONF) and for the 1.5B models. A learned target transferred across families is worse everywhere. Only for Qwen2.5-Math-7B with large budgets is there a small gain.

All code, tables and the generator of every number in this paper are released.[1]

2 Related Work

Process supervision. Step-level supervision outperforms outcome supervision [2, 3]; Monte Carlo rollouts can label steps automatically [4], but such labels disagree with human judgements because a completion can reach the right answer from a wrong step and vice versa, which Zhang et al. [6] treat as label noise. ProcessBench [5] evaluates first-error identification. We study the disagreement itself.

Predicting failure early. Models carry information about their own correctness [22, 21], and hidden states predict CoT success before or during generation [9, 10]; conformal methods certify reasoning prefixes [20]. Bulut [11] shows that such pooled AUROCs are largely explained by problem difficulty and fall to chance within a problem. We apply the same control to text detectors on human-annotated traces.

Which steps matter. Resampling from prefixes identifies influential sentences [12], tokens [13, 14] and the importance of the first step [15]. Merrill and Srivastava [16] localise the point of no return for deceptive behaviour and name incorrect solutions as future work; recoverability of error states has been studied for distillation [17]. None compares a counterfactual decision point with human first-error labels. Work on performative CoT [18, 19] finds that answers are often fixed apart from the written reasoning, which our non-fatal errors echo.

Correcting and backtracking. Unaided self-correction rarely helps [8], but models can correct a mistake once told where it is [7], which motivates verifier-guided search that returns to the first error. Independent resampling from scratch, as in self-consistency [23], is the natural baseline for any such scheme; we compare the two directly. Compute-matched comparisons of verifier-guided search with independent sampling are known to depend on difficulty and budget [29]; we add counterfactual ground truth on where the restart point should be.

3 Data and Definitions

Data. ProcessBench [5] contains 3,400 step-segmented solutions to GSM8K, MATH, OlympiadBench and Omni-MATH problems (2,842 distinct problems) from 12 open models, each with the index e of the first erroneous step (or none) and whether the final answer is correct. Correct and incorrect answers are balanced within each tier by construction.

Outcome groups. A trace is *clean* (no annotated error; 1,179), *recovered* (an error, but a correct answer; 521) or *fatal* (an error and a wrong answer; 1,700). No clean trace has a wrong answer. The *fatality* of first errors is $P(\text{wrong} \mid \text{error})$.

Counterfactual value. For a trace $T = (s_1, \ldots, s_n)$ written by model M, let $v_k = P_M(\text{correct} \mid q, s_{1:k})$, estimated by sampling m continuations of the prefix from M; v_0 is M's from-scratch solve rate and v_n is the observed outcome. The *decisive step* is $d = \arg\max_k(v_{k-1} - v_k)$, and the *point of no return* is $c = \min\{k : v_j \leq \delta \; \forall j \geq k\}$. A failing trace is *capability-limited* if $v_0 \leq \delta$, shows *latent commitment* if $c < e$, *commits at the error* if $c = e$, and has a *recoverable error* if $c > e$.

Evidence levels. The hypotheses H1–H4 of the counterfactual experiment (the commitment categories, the decisive step, sample-split backtracking, $\delta = 1/8$ with 1/16 and 1/4 as sensitivity) were committed before any rollout. Analyses added after seeing the Run A/B data (restart vs. first error, repeated splitting, the position-matched null) are *exploratory*. Before the confirmatory run (Run C) we committed a second plan (git commit `459ff22`; `docs/PREREGISTRATION_RUN_C.md`). It specified:

- tests C1–C4 on Llama-3.1-8B, a new family whose traces cover 479 new questions;
- measurement checks R1–R3 on three new independent sampling batches;
- the practical policies and their cost accounting.

Run C reached 144 of 274 Llama fatal traces within its 11.5-hour limit. Traces were processed in a random order, and recovered and clean traces were scheduled after the fatal ones, so none of those were run. All Run C deviations are reported in the text.

Statistics. Detectors are TF-IDF (uni+bigrams) or one-hot features with logistic regression [28], cross-fitted over 5 folds grouped by problem and repeated over 5 seeds. Intervals are 95% question-level cluster bootstrap intervals [27] (1,000 resamples); differences between detectors use a paired bootstrap on the same traces.

4 Observational Results

4.1 Early failure prediction is a difficulty signal

Table 1 and Fig. 1 compare detectors that read the question, the generator and tier, the reasoning prefix, or combinations. The question alone reaches 0.561 and question plus generator 0.588 [0.570, 0.607]. A detector reading only the first 25% of the reasoning is *worse* than that (-0.025, $p = 0.012$), and adding the 25% prefix to question and generator changes AUROC by +0.005 [-0.004, +0.014]. The reasoning adds +0.022 at 50% and +0.031 [+0.020, +0.041] at 100%, consistently across all five seeds.

Within a problem the generator is the only varying factor besides the reasoning: over 326 problems with both a failing

[1] `https://github.com/HarshDubey23/early-failure-detection-cot`

Table 1: AUROC for a wrong final answer (out-of-fold, 3,400 traces). Q: question text; G: generator and tier.

Detector	AUROC	95% CI	SD (5 seeds)
Q	0.561	0.543–0.578	0.005
G	0.539	0.521–0.557	0.002
Q+G	0.588	0.570–0.607	0.003
Prefix 25%	0.564	0.545–0.582	0.006
Prefix 100%	0.600	0.582–0.619	0.004
Q+G+prefix 25%	0.593	0.575–0.611	0.004
Q+G+prefix 50%	0.610	0.592–0.628	0.004
Q+G+prefix 100%	0.619	0.601–0.637	0.003
Trace length	0.542	0.524–0.561	0.002

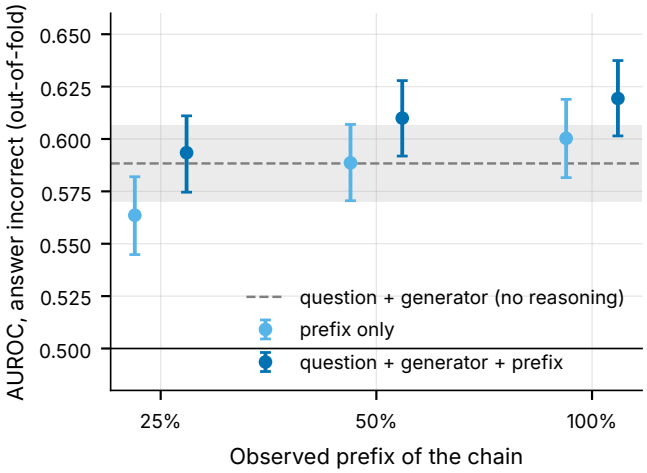


Figure 1: The prefix adds nothing beyond question and generator until half the chain is observed.

and a succeeding trace (466 pairs), the 25% prefix orders pairs correctly 0.485 of the time [0.440, 0.536] and the full prefix 0.539 [0.487, 0.589], neither better than generator identity alone (0.548). Early failure prediction on this benchmark is therefore almost entirely a statement about which problems are hard for which models.

4.2 One first error in four is not fatal

Fatality of first errors is 0.765 [0.748, 0.782]: 23.5% of annotated first errors are followed by a correct answer (Fig. 2). Fatality falls with difficulty, from 0.966 on GSM8K to 0.659 on Omni-MATH, and is flat in the position of the error (0.749–0.766 across onset quartiles; 0.890 only when the error is in the final step). Fatal and recovered errors occur at the same relative position (Mann–Whitney $p = 0.21$, $r = 0.04$).

Explicit re-checking language after the error is barely more frequent in recovered than in fatal traces (18.6% vs 15.5%, $p = 0.09$), so self-correction does not explain most recoveries. Human coding of a stratified sample of recovered traces by two independent coders is in progress (protocol released); Section 5 measures the same question without coders, by forcing the model to answer at every step.

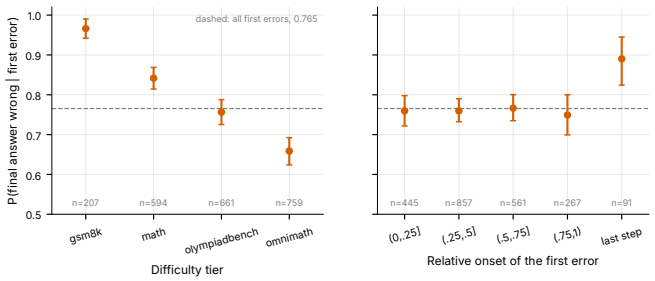


Figure 2: Fatality of first errors by difficulty tier and by where the error occurs.

Table 2: AUROC between outcome groups (positive vs negative).

Detector (trained on)	rec. vs clean	fatal vs rec.	fatal vs clean
Prefix 100% (wrong answer)	0.703	0.456	0.664
Q+G+prefix (wrong answer)	0.679	0.501	0.672
Prefix 100% (has error)	0.819	0.376	0.709

4.3 Detectors track errors, not commitment

If a failure detector captured commitment, it would score recovered traces like clean ones. It does the opposite (Table 2): trained to predict a wrong answer, the full-prefix detector separates recovered from clean traces (0.703) but ranks fatal traces *below* recovered ones (0.456 [0.428, 0.484]); with question and generator included the ranking is at chance (0.501). A detector trained to find errors inverts more strongly (0.376). Among erroneous traces, the fatality of the first error is predicted by question and generator (AUROC 0.671), and the error step, the prefix up to it or the next one or two steps change this by at most 0.003.

5 Counterfactual Commitment

The observational results show that first errors and outcomes come apart. They do not show where a failing trajectory actually becomes unrecoverable. We measure that directly by resampling.

5.1 Setup

Generators and traces. We use four ProcessBench generators whose weights fit on two 16 GB GPUs, from two model families:

- Qwen2.5-1.5B-Instruct and Qwen2-1.5B-Instruct [24]: all their traces (116 and 88).
- Qwen2.5-Math-7B-Instruct [25]: all fatal and recovered traces with a gold answer, and 60 clean traces (314 in total).
- Llama-3.1-8B-Instruct: the fatal traces completed within the time limit of the confirmatory run (144; Section 3).

The Llama traces cover 512 questions, 479 of which appear in none of the Qwen traces (1,069 distinct questions in total). Gold answers come from the source datasets, matched on problem text, and are compared with `math-verify`.

Rollouts. For every prefix $s_{1:k}$, $k = 0, \ldots, n-1$, we sample m continuations from the trace's own generator with vLLM [26] in float16, at temperature 0.7 and top-p 0.95, and estimate v_k as the fraction that reach the gold answer. The sample sizes are:

- the 1.5B models: $m = 64$, as four independent sampling batches of 16 with different seeds;
- Qwen2.5-Math-7B: $m = 12$;
- Llama: $m = 16$.

Continuations are capped at 1,536 new tokens for the 1.5B models and at 1,024 for the larger models. At the cap, 17% (Qwen2.5-Math-7B) and 29% (Llama) of continuations end without an answer. In total we score 125,832 continuations. Every continuation, its answer and its correctness are released.

What can and cannot be read off v_k. Under the generating policy, v_k is a martingale. Curves conditioned on the eventual outcome therefore drift even when no single step matters: for any fixed k (and any stopping time),

$$\mathbb{E}[v_k \mid \text{fail}] = \frac{\mathbb{E}[v_k(1-v_k)]}{P(\text{fail})} \leq v_0, \qquad (1)$$

because $x(1-x)$ is concave (Appendix A). We therefore never interpret a downward drift of failing curves by itself. We ask whether the drop is *concentrated* at the annotated error, against a position-matched null and the noise floor of clean traces. We evaluate backtracking targets with sample splitting.

5.2 How precise are the value curves?

Each v_k is a binomial estimate, and the decisive step $d = \arg\max_k(v_{k-1} - v_k)$ is a noisy argmax. We quantify both kinds of uncertainty (Fig. 3).

Value curves. With the four independent batches of the 1.5B models split two against two, v_k correlates at 0.985 and 0.976 across all prefixes (Spearman–Brown projection to $m = 64$: 0.993 and 0.988). The random split-half correlation is 0.951 for Qwen2.5-Math-7B ($m = 12$) and 0.896 for Llama ($m = 16$).

Decisive steps. We draw every v_k from its Jeffreys posterior and recompute d 2,000 times. The modal step has posterior probability at least 0.5 in 94%, 100%, 87% and 88% of the not-capability-limited fatal traces. Independent halves of the continuations pick the same decisive step in 0.765, 0.906, 0.648 and 0.533 of traces, against position-matched nulls of 0.156–0.284. The posterior probability that the decisive step is the annotated error is bimodal: for most traces it is either clearly yes or clearly no. Disagreement between decisive steps and labels is therefore not measurement noise.

5.3 Where failing trajectories commit

Table 3 classifies every fatal trace at $\delta = 1/8$.

Many failures are decided before the first step. 33%, 52%, 66% and 58% of fatal traces are capability-limited: the model solves the problem from scratch at most one time in eight, so no step of the trace is a choke point. This is the counterfactual counterpart of Section 4's difficulty result.

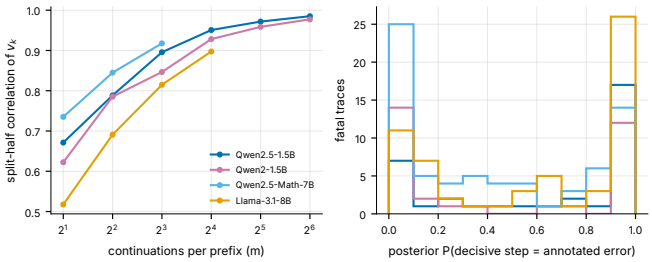


Figure 3: Left: split-half correlation of v_k as a function of the continuations per prefix. Right: posterior probability, per fatal trace, that the decisive step is the annotated first error.

Table 3: Where fatal traces commit ($\delta = 1/8$). "Decisive": share of not-capability-limited fatal traces whose largest drop in v is at the annotated first error; null: position-matched. Llama: fatal traces only.

	Qwen2.5-1.5B	Qwen2-1.5B	Qwen2.5-Math-7B	Llama-3.1-8B
Traces (clean/fatal/rec.)	53/51/12	15/67/6	60/207/47	–/144/–
Continuations per prefix	64	64	12	16
Capability-limited	0.333	0.522	0.657	0.583
Latent commitment ($c < e$)	0.137	0.119	0.063	0.076
Commits at error ($c = e$)	0.471	0.254	0.179	0.208
Recoverable error ($c > e$)	0.059	0.104	0.101	0.132
Decisive = first error	0.647	0.438	0.423	0.633
position-matched null	0.166	0.204	0.159	0.187
n	34	32	71	60
Drop at error: fatal	+0.251	+0.162	+0.109	+0.179
Drop at error: recovered	-0.129	+0.211	-0.215	–
Noise floor (clean)	+0.058	+0.201	+0.078	–

When the outcome is still open, the annotated error is often where it closes. Among fatal traces that are not capability-limited, the first error is the largest drop in v in 0.647, 0.438, 0.423 and 0.633 of traces. The position-matched nulls are 0.166, 0.204, 0.159 and 0.187. The differences are +0.481 [+0.330, +0.627], +0.234 [+0.063, +0.407], +0.264 [+0.154, +0.382] and +0.447 [+0.328, +0.569]. For Llama this was preregistered (C3) and is confirmed. The remaining traces show one of two patterns:

- *latent commitment*: success has already collapsed before the written error (21%, 25%, 18% and 18% of not-capability-limited fatal traces);
- *recoverable errors*: success stays above δ past the error (9%, 22%, 30% and 32%).

Event study. Aligned at the annotated error (Fig. 4), fatal traces show a step change from $r = -1$ to $r = 0$: 0.280→0.029, 0.208→0.046, 0.175→0.066 and 0.241→0.062. Clean traces aligned at a random step drift smoothly upward, as the success-conditioned analogue of Eq. 1 predicts. For recovered traces of Qwen2.5-Math-7B, success *rises* across the annotated error (0.374→0.589), by far more than the drift of clean traces.

Does the model already know its answer? (exploratory) At every prefix we force an immediate answer (65,976 forced answers), giving an *answer curve* a_k. The probe is valid at the endpoints. After the full trace, the forced answer reproduces the trace's own answer in 79–91% of cases for three models, but in only 47% for Qwen2-1.5B, for which the probe is unreliable. From the bare question it is right in 11–14%

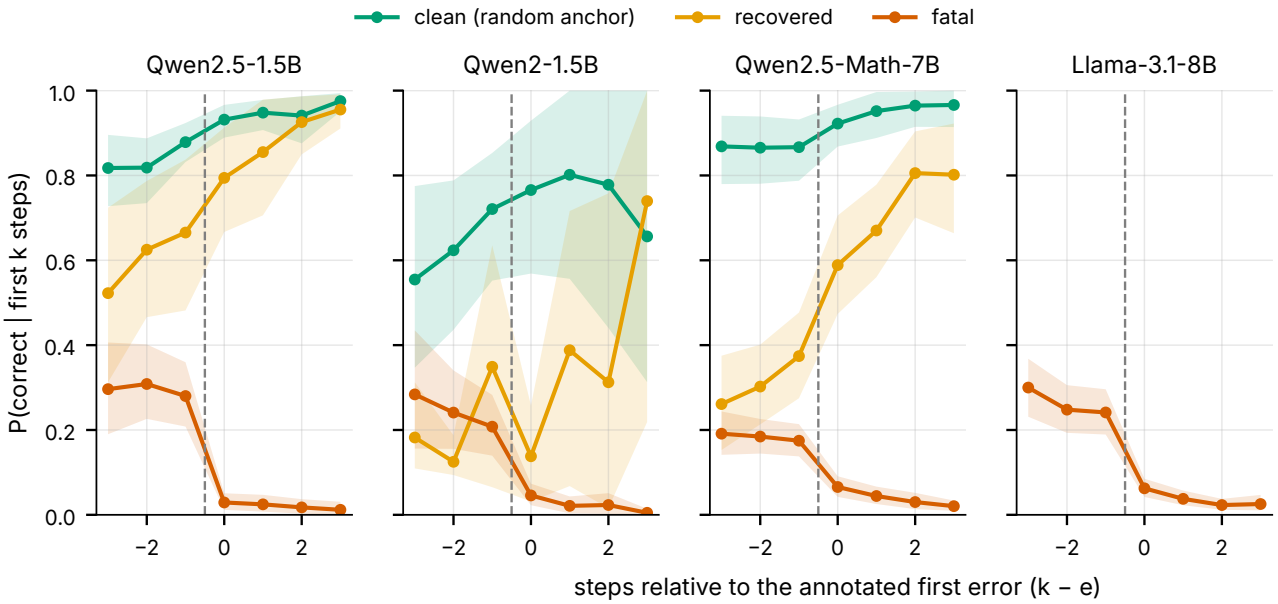


Figure 4: Success probability v_k aligned at the annotated first error ($k - e$; the dashed line separates the last correct step from the error). Shaded: 95% bootstrap CI. Clean traces are aligned at a uniformly drawn step (placebo). Llama: fatal traces only.

of cases. Answers become accessible late, mostly after the error position (Fig. 5). In fatal traces, the model's accessible answer was correct just before the error and wrong just after it in only 0.058 (Qwen2.5-Math-7B) and 0.107 (Llama) of traces. In most fatal traces the forced answer was never right (0.850 and 0.824). In recovered traces, the correct answer is already accessible right after the error in 0.340 ($n = 47$) and 0.268 ($n = 97$) of traces. This is a lower bound on recoveries in which the written error did not matter, because answers are computed late even in clean traces.

5.4 Where to backtrack, per attempt

Verifier-guided search resamples a failing trace from some earlier point. We compare four targets on fatal traces:

- restarting from scratch (v_0);
- resampling from just before the annotated first error (v_{e-1});
- resampling from just before the decisive step (v_{d-1});
- the best prefix in hindsight (an in-sample upper bound).

Choosing a step and scoring it on the same continuations would reward noise, so we split every prefix's continuations at random into halves A and B. Half A selects the trace ($v_0^A > \delta$) and its decisive step, and half B scores every target. We average over 50 splits and bootstrap over traces. On a planted-effect simulation in which the first error is the decisive step by construction, this estimator returns $+0.001$, so it does not create a gap.

For the three Qwen models, backtracking to the annotated first error is worse than starting over. This was found exploratorily in Run B. Restart beats first-error backtracking by +0.123, +0.165 and +0.068 (Table 4). On the 48 new continuations per prefix alone, which were sampled after the effect was found, it replicates for both 1.5B models. The pooled difference is +0.140 [+0.078, +0.202] (preregistered R1, confirmed). The steps before the error already lower the chance of success. Resampling from just before the decisive step recovers the loss: it beats first-error backtracking by +0.103, +0.147 and +0.108 (all CIs above zero).

For Llama-3.1-8B, it is not. The preregistered confirmatory tests on the new model family fail:

- restart − first error is -0.022 [-0.090, +0.047] (C1, Holm-adjusted $p = 1.00$);
- decisive step − first error is +0.010 [-0.030, +0.058] (C2, Holm-adjusted $p = 1.00$).

This is not an artefact of truncation. With continuations that hit the token limit treated as missing rather than wrong, restart − first error is -0.007 [-0.068, +0.048]. The Llama result is consistent with its localization result: the annotated error is the decisive step more often (0.633), so the step before the error is usually still a good place to resume. Backtracking to human first errors is therefore model-dependent. It is worse than restarting for three Qwen models, but not in general.

Decisive-step backtracking and restart: no difference detected, equivalence mostly untested. Decisive − restart is within ± 0.04 for every model, and every 95% CI includes zero. Equivalence at a margin of ± 0.05 (TOST, 90% CI) holds for Qwen2.5-1.5B and for the pooled new batches. It is inconclusive for the other models, including the preregistered C4 test on Llama. We therefore do not claim that decisive-step backtracking matches restart; we claim only that no difference was detected. It uses 15–29% fewer continuation tokens per attempt. The best prefix in hindsight reaches 0.549–0.678.

Reading. Eq. 1 explains why restart is a strong baseline: for a backtracking point at a fixed depth or at a stopping time of the trace, failed on-policy prefixes are on average no better than the empty prefix. The point just before the first error is

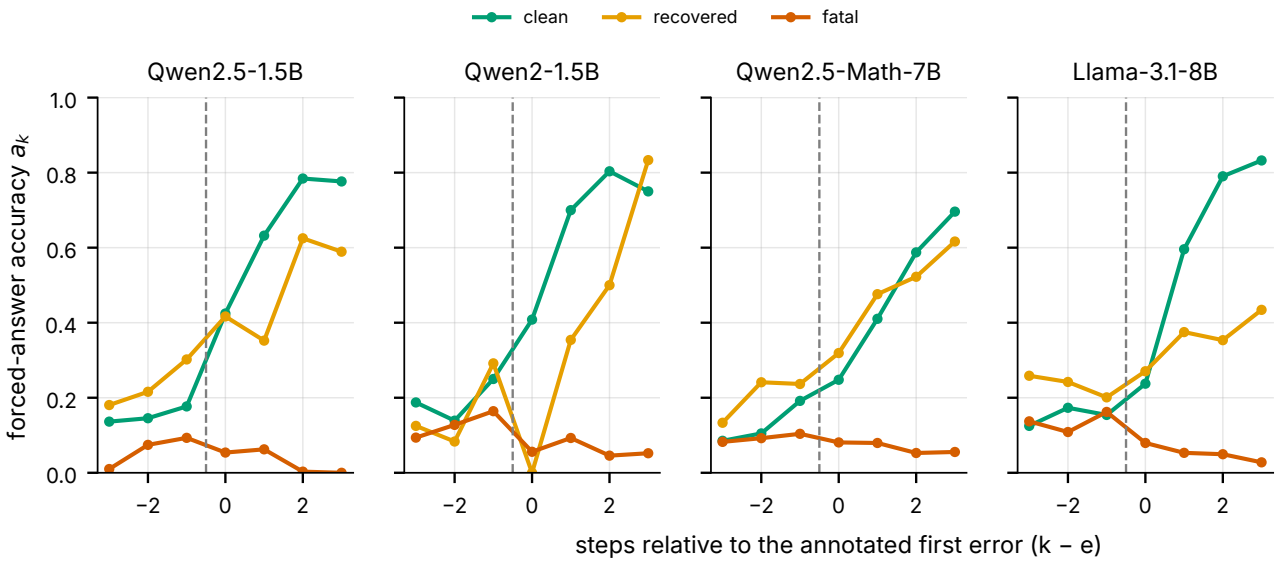


Figure 5: Answer curves: accuracy of an immediately forced answer after k steps, aligned at the annotated first error (clean traces at their middle step).

Table 4: Success after resampling a fatal trace from each target, per attempt (sample-split, 50 splits; mean pool size in parentheses). 95% trace-bootstrap CIs. Evidence level per row: see Section 3.

	Qwen2.5-1.5B (34)	Qwen2-1.5B (32)	Qwen2.5-Math-7B (76)	Llama-3.1-8B (56)
Restart	0.501	0.575	0.470	0.537
Before annotated first error	0.378	0.410	0.402	0.559
Before decisive step	0.481	0.557	0.511	0.569
Best prefix (oracle, in-sample)	0.549	0.621	0.622	0.678
Restart − first error	+0.123 [+0.054, +0.197]	+0.165 [+0.067, +0.265]	+0.068 [+0.007, +0.133]	-0.022 [-0.090, +0.047]
new batches only (replication)	+0.128 [+0.055, +0.206]	+0.151 [+0.054, +0.249]	–	–
truncated continuations dropped	–	–	+0.107 [+0.044, +0.176]	-0.007 [-0.068, +0.048]
Decisive − first error	+0.103 [+0.053, +0.161]	+0.147 [+0.061, +0.236]	+0.108 [+0.051, +0.165]	+0.010 [-0.030, +0.058]
Decisive − restart	-0.020 [-0.052, +0.013]	-0.018 [-0.061, +0.032]	+0.040 [-0.008, +0.089]	+0.032 [-0.017, +0.085]
TOST at ±0.05 (90% CI)	equivalent	inconclusive	inconclusive	inconclusive
Continuation tokens: restart / decisive	376 / 289	210 / 178	650 / 463	430 / 316

neither, because it depends on the next step. So the inequality allows first-error backtracking to beat restart or to lose to it. The data show both outcomes, depending on the model.

6 Practical Policies at a Matched Budget

Section 5.4 compares targets per attempt, using oracle knowledge of the decisive step. A deployed system has to choose a restart point without rollouts and pays for every token. We therefore compare practical policies against restart under a matched end-to-end budget.

Policies. All were fixed before the confirmatory data existed (Section 3).

- Restart.
- PRM first error: resample from before the first step that Qwen2.5-Math-PRM-7B [6] scores below 0.5.
- PRM largest drop: resample from before the step with the largest fall in PRM score.
- Probe search: two probe continuations at each of three prefixes, then resample from the latest prefix with a success.
- Learned: logistic regression on step features (PRM score and drop, the generator's NLL and entropy, position, length), trained to predict the decisive step on the *other* generators. For Llama, it is trained only on the three Qwen models.
- Two references: human first error (the expert label, not available in deployment) and the oracle decisive step (all its rollouts counted).

Implementation check. Our implementation of the verifier loads the reward head directly from the checkpoint. It reaches a ProcessBench F1 of 73.2 over 3,374 traces (GSM8K 81.8, MATH 77.4, OlympiadBench 67.0, Omni-MATH 66.7). The published value is 73.5 [6].

Cost.

- Tokens are weighted by model size relative to the generator

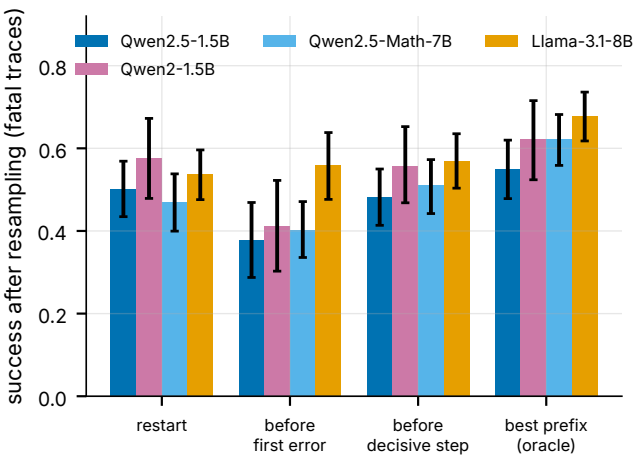


Figure 6: Success after resampling fatal traces from each target (half-B estimates, 50 splits).

(FLOPs $\propto$ parameters $\times$ tokens).

- Every prefill (question plus prefix) counts once per restart point, assuming prefix caching.
- Every generated token counts.
- Verifier passes, probes and oracle rollouts are charged to the policies that use them.
- The budget is b times the cost of one restart attempt, for each trace.
- A policy spends what is left on independent attempts from its restart point. We report the probability that at least one attempt is correct (Jeffreys-posterior estimate from held-out continuations, 50 splits, trace bootstrap).

No practical policy beats restart reliably (Table 5, Fig. 7).

- **Llama (confirmatory).** PRM largest-drop backtracking is *worse* than restart at four attempts, -0.063 [-0.100, -0.024]. So is the learned policy transferred from Qwen, at -0.187 [-0.226, -0.148].
- **The 1.5B models.** Scoring the trace with a 7B verifier costs about five generator attempts, so every verifier-based policy loses heavily at small budgets.
- **Qwen2.5-Math-7B.** PRM largest-drop backtracking is the only practical policy with a positive estimate. Under full accounting it is -0.022 [-0.052, +0.010] at four attempts and +0.030 [+0.003, +0.057] at eight. If the verifier pass is assumed to be paid anyway, it is +0.046 [+0.017, +0.075] (exploratory).
- **Probe search and the oracle.** Both spend most of the budget on finding the restart point, and lose.

Even a perfect first-error verifier is no shortcut. The human first error, which no deployed system has, is within a few points of restart at matched compute for every model: -0.047, -0.075, +0.039 and +0.001. Later restart points buy more attempts, but each attempt is worth less, which is the trade-off the proposition in Appendix A anticipates.

7 Discussion

For early warning. Reported AUROCs for early failure prediction should come with a question-only baseline, a generator baseline, the increment over both, and a within-problem concordance. Without them, a detector of problem difficulty cannot be told apart from a detector of reasoning failure. On ProcessBench, text detectors pass this test only after half the chain. The counterfactual measurement explains why. Between 33% and 66% of failures are fixed before the first step, so reading the trajectory can add little for them.

For process supervision. Human first-error labels mark where reasoning first goes wrong, which is the right target for auditing. They are also, more often than chance, where success collapses: 0.423–0.647 of open failures, confirmed on a new model family. But using them as restart points depends on the model. Per attempt it was worse than restarting for three Qwen models and equal for Llama, and at matched compute it never clearly beat restarting. Verifier-guided backtracking should therefore always be reported against restart at matched end-to-end compute, with the verifier's own cost included.

What would change the picture. The oracle shows headroom: the best restart point in hindsight succeeds 0.549–0.678 per attempt, against 0.470–0.575 for restart. A verifier that predicted the decisive step cheaply and accurately could exploit that headroom. Our learned predictor does not: transferred across families, it is worse than restart. The released value curves are labelled training data for such a predictor.

8 Limitations and what remains unverified

- **Coverage.**
  - Four open models of at most 8B parameters, mathematics only.
  - Llama contributes 144 fatal traces. Its recovered and clean traces were not run, so Llama has no noise floor, no event study for those groups, and no answer-curve comparison by rollout.
  - Qwen2.5-7B-Instruct and larger models were not run.
- **On-policy approximation.** ProcessBench re-segmented the original solutions, and the decoding settings that produced them are unknown. Prefixes are therefore near-, not exactly, on-policy, and the proposition holds only approximately for them.
- **Truncation.** For the larger models, 17% and 29% of continuations hit the token cap without an answer. Treating them as missing changes no conclusion, but it lowers absolute success rates.
- **Equivalence.** Equivalence of decisive-step backtracking and restart at ± 0.05 is established only for Qwen2.5-1.5B and the pooled new batches. Elsewhere, including the preregistered Llama test, it is inconclusive.
- **Exploratory results.** The difficulty and fatality analyses, the answer curves, the shared-verifier accounting and all policy results on the Qwen models are exploratory.

Table 5: P(solved) minus restart for fatal traces at a budget of four restart attempts (FLOP-matched, all overheads charged). 95% trace-bootstrap CIs. "Shared verifier": the PRM pass is not charged, as if it had been run anyway to detect the failure (exploratory). Llama rows are confirmatory for PRM largest drop and Learned; all other rows are exploratory.

	Qwen2.5-1.5B	Qwen2-1.5B	Qwen2.5-Math-7B	Llama-3.1-8B
Restart: P(solved)	0.657	0.514	0.437	0.492
PRM first error	-0.651 [-0.744, -0.558]	-0.514 [-0.603, -0.424]	-0.041 [-0.066, -0.016]	-0.068 [-0.102, -0.035]
PRM largest drop	-0.651 [-0.744, -0.558]	-0.514 [-0.603, -0.424]	-0.022 [-0.052, +0.010]	-0.063 [-0.100, -0.024]
Probe search	-0.646 [-0.739, -0.554]	-0.448 [-0.537, -0.361]	-0.428 [-0.468, -0.388]	-0.489 [-0.539, -0.436]
Learned	-0.657 [-0.749, -0.563]	-0.514 [-0.603, -0.424]	-0.151 [-0.177, -0.127]	-0.187 [-0.226, -0.148]
Human first error (reference)	-0.047 [-0.122, +0.027]	-0.075 [-0.132, -0.022]	+0.039 [+0.012, +0.066]	+0.001 [-0.025, +0.029]
Oracle decisive step, rollouts counted	-0.657	-0.514	-0.437	-0.492
PRM largest drop, shared verifier	-0.061 [-0.146, +0.022]	-0.098 [-0.160, -0.040]	+0.046 [+0.017, +0.075]	+0.003 [-0.030, +0.036]

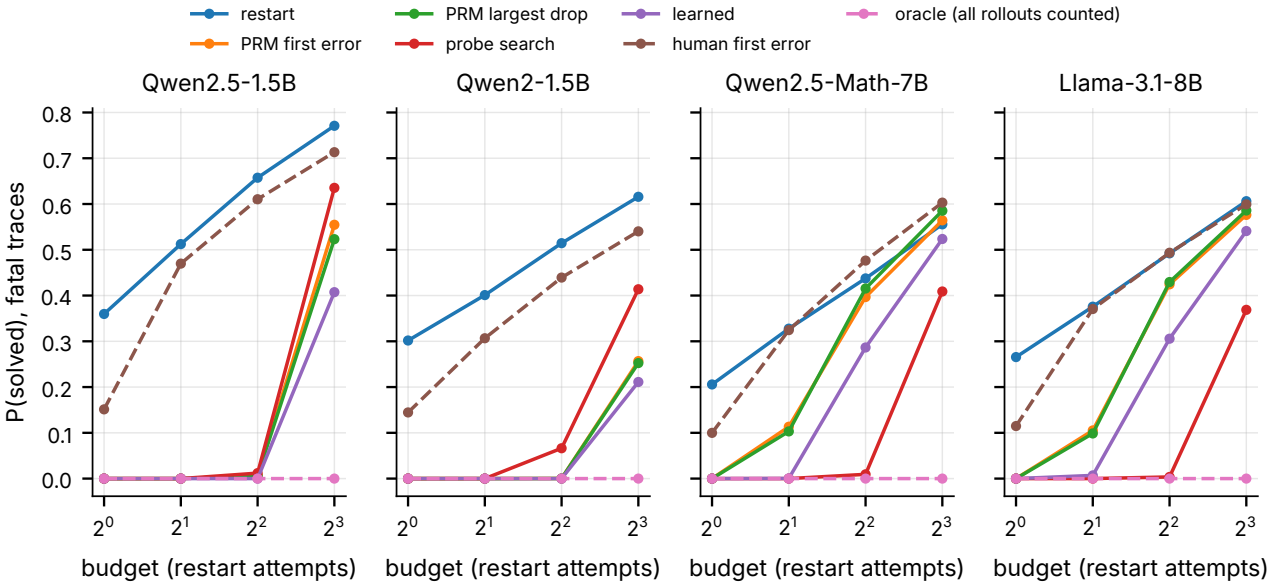


Figure 7: P(solved) for fatal traces against the end-to-end budget (in restart attempts, FLOP-matched, all overheads charged).

- **Human coding.** The coding of recovered traces by two independent human coders has not been done. The protocol and blinded sheets are released, and no coding result is reported.

9 Conclusion

Errors are where reasoning first goes wrong; choke points are where it stops being recoverable. Measured counterfactually, on two model families with independent replication, three things hold. First, many failures are decided before the first step. Second, when the outcome is still open, the human-labelled error is a better-than-chance but far from perfect locator of the choke point. Third, neither human labels nor a strong process reward model provide a restart point that beats simply starting over at matched compute. Restart is the baseline that verifier-guided backtracking has to beat, and the counterfactual value curve is the ground truth for where a restart point should be.

References

[1] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *NeurIPS*, 2022.

[2] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe, "Let's verify step by step," arXiv:2305.20050, 2023.

[3] J. Uesato *et al.*, "Solving math word problems with process- and outcome-based feedback," arXiv:2211.14275, 2022.

[4] P. Wang *et al.*, "Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations," in *ACL*, 2024.

[5] C. Zheng, Z. Zhang, B. Zhang, R. Lin, K. Lu, B. Yu, D. Liu, J. Zhou, and J. Lin, "ProcessBench: Identifying process errors in mathematical reasoning," in *ACL*, 2025.

[6] Z. Zhang *et al.*, "The lessons of developing process reward models in mathematical reasoning," arXiv:2501.07301, 2025.

[7] G. Tyen, H. Mansoor, V. Cărbune, P. Chen, and T. Mak, "LLMs cannot find reasoning errors, but can correct them given the error location," in *Findings of ACL*, 2024.

[8] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in *ICLR*, 2024.

[9] A. Afzal, F. Matthes, G. Chechik, and Y. Ziser, "Knowing before saying: LLM representations encode information about chain-of-thought success before completion," arXiv:2505.24362, 2025.

[10] W. Lugoloobi, T. Foster, W. Bankes, and C. Russell, "LLMs encode their failures: Predicting success from pre-generation activations," arXiv:2602.09924, 2026.

[11] Y. U. Bulut, "It's the problem, not the path: Budget and difficulty confounds in LLM reasoning trajectories," arXiv:2609.03436, 2026.

[12] P. C. Bogdan, U. Macar, N. Nanda, and A. Conmy, "Thought anchors: Which LLM reasoning steps matter?" arXiv:2506.19143, 2025.

[13] Z. Lin *et al.*, "Critical tokens matter: Token-level contrastive estimation enhances LLM's reasoning capability," in *ICML*, 2025.

[14] E. Bigelow, A. Holtzman, H. Tanaka, and T. Ullman, "Forking paths in neural text generation," in *ICLR*, 2025.

[15] B. Liao *et al.*, "Lost at the beginning of reasoning," arXiv:2506.22058, 2025.

[16] S. Merrill and S. Srivastava, "The point of no return: Counterfactual localization of deceptive commitment in language-model reasoning," arXiv:2605.17113, 2026.

[17] D. Jiang, Z. Zhang, K. Yuan, and S. Ma, "Not every divergence should be suppressed: Counterfactual recoverability in on-policy distillation," arXiv:2608.04408, 2026.

[18] W. Li, F. Yang, A. Hazarika, S. A. Mehta, and K. Onoue, "When reasoning traces become performative: Step-level evidence that chain-of-thought is an imperfect oversight channel," arXiv:2605.11746, 2026.

[19] S. Boppana *et al.*, "Reasoning theater: Disentangling model beliefs from chain-of-thought," arXiv:2603.05488, 2026.

[20] M. Y. Cheung, A. Veeraraghavan, H. Chen, and G. Balakrishnan, "Conformal certification of reasoning trace prefixes," arXiv:2605.30085, 2026.

[21] H. Orgad *et al.*, "LLMs know more than they show: On the intrinsic representation of LLM hallucinations," arXiv:2410.02707, 2024.

[22] S. Kadavath *et al.*, "Language models (mostly) know what they know," arXiv:2207.05221, 2022.

[23] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in *ICLR*, 2023.

[24] A. Yang *et al.*, "Qwen2.5 technical report," arXiv:2412.15115, 2024.

[25] A. Yang *et al.*, "Qwen2.5-Math technical report: Toward mathematical expert model via self-improvement," arXiv:2409.12122, 2024.

[26] W. Kwon *et al.*, "Efficient memory management for large language model serving with PagedAttention," in *SOSP*, 2023.

[27] C. A. Field and A. H. Welsh, "Bootstrapping clustered data," *J. R. Stat. Soc. B*, vol. 69, no. 3, pp. 369–390, 2007.

[28] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[29] C. Snell, J. Lee, K. Xu, and A. Kumar, "Scaling LLM test-time compute optimally can be more effective than scaling model parameters," arXiv:2408.03314, 2024.

A Why restarting is a strong baseline

Let a trace $s_1, \ldots, s_n$ be sampled from policy M for question q, let S be the event that its final answer is correct, $\mathcal{F}_k = \sigma(s_{1:k})$, and $v_k = P_M(S \mid q, \mathcal{F}_k)$. Resampling a fresh continuation of $s_{1:k}$ from M succeeds with probability v_k.

Proposition. For every stopping time $\tau \leq n$ of $(\mathcal{F}_k)$, $\mathbb{E}[v_\tau \mid \neg S] \leq v_0$ (whenever $v_0 < 1$).

Proof. (v_k) is a bounded martingale and $g(x) = x(1-x)$ is concave, so $g(v_k)$ is a bounded supermartingale; by optional stopping $\mathbb{E}[g(v_\tau)] \leq g(v_0)$. By the tower property, $\mathbb{E}[v_\tau \mathbf{1}_{\neg S}] = \mathbb{E}[v_\tau(1-v_\tau)] = \mathbb{E}[g(v_\tau)]$. Dividing by $P(\neg S) = 1 - v_0$ gives $\mathbb{E}[v_\tau \mid \neg S] \leq v_0(1-v_0)/(1-v_0) = v_0$. □

Consequences. (i) Any backtracking rule that picks its restart point by reading the failed trace only up to that point (a fixed depth, a confidence threshold crossed so far, a verifier that flags step k using $s_{1:k}$ and resamples from k) cannot, per attempt and in expectation over failed on-policy traces, beat restarting from scratch. (ii) Rules that look ahead can: resampling from *before* a flagged step uses step $k+1$, and the decisive step uses rollouts. The proposition therefore sets the bar a backtracking target must clear, and the paper measures how far human first errors and verifier signals clear it. (iii) The proposition is per attempt. Under a fixed compute budget a later restart point buys more attempts, because continuations are shorter, which is why Section 6 compares policies at a matched end-to-end budget. (iv) ProcessBench traces were not necessarily sampled at the temperature used for v_k, so the proposition holds only approximately for them; the empirical comparisons do not rely on it.

B Preregistered tests and outcomes

Test	Hypothesis (data)	Estimate [95% CI]	Outcome
H1	first error is the decisive step more often than chance (Qwen, Run B)	see Table 3	supported (Qwen2-1.5B CI includes null at $m = 16$)
C1	restart > first-error backtracking (Llama)	-0.022 [-0.090, +0.047]	not confirmed
C2	decisive-step > first-error backtracking (Llama)	+0.010 [-0.030, +0.058]	not confirmed
C3	decisive share > position-matched null (Llama)	+0.447 [+0.328, +0.569]	confirmed (Holm-adjusted p <0.002)
C4	decisive-step ≡ restart within ± 0.05 (Llama)	90%: [-0.008, +0.076]	inconclusive
R1	restart > first error on new batches only (Qwen 1.5B, pooled)	+0.140 [+0.078, +0.202]	confirmed
R2	v_k reliable across independent batches	$r = 0.985$, 0.976	confirmed
R3	decisive step stable under posterior noise	modal ≥ 0.5: 94%, 100%	confirmed
P2, P4	practical policies vs restart at $b = 4$ (Llama)	-0.063, -0.187	both worse than restart

C Threshold and definition sensitivity

	Qwen2.5-1.5B	Qwen2-1.5B	Qwen2.5-Math-7B	Llama-3.1-8B
Restart − first error, $\delta = 1/16$	+0.112	+0.155	+0.068	-0.011
Restart − first error, $\delta = 1/8$	+0.123	+0.165	+0.068	-0.022
Restart − first error, $\delta = 1/4$	+0.136	+0.186	+0.116	-0.031
First error = largest drop, $\delta = 1/16$ / $1/8$ / $1/4$	0.561 / 0.647 / 0.633	0.405 / 0.438 / 0.481	0.386 / 0.423 / 0.404	0.621 / 0.633 / 0.706
First error = point of no return	0.634 / 0.706 / 0.600	0.568 / 0.531 / 0.556	0.477 / 0.521 / 0.489	0.455 / 0.500 / 0.588
First error = first drop above clean floor	0.390 / 0.471 / 0.467	0.216 / 0.250 / 0.296	0.205 / 0.239 / 0.277	–

Restart − first error keeps its sign and significance at every threshold for every model: positive for the three Qwen models and null for Llama. The clean-floor definition is unavailable for Llama (no clean traces were run).